

No need to show the parse tree, included for better understanding

ID: Obtained Marks

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$$\begin{aligned} S &\rightarrow OAB \\ A &\rightarrow 00A|01A|10A|11A|E \\ B &\rightarrow 011 \end{aligned} \quad O(\Sigma\Sigma)^*(O+1)$$

b) Every second letter in ω is b

$$((a+b)b)^*(\epsilon+ab)(\epsilon+a+b)$$

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow XA|E \\ X &\rightarrow PQ \\ P &\rightarrow a|b \\ Q &\rightarrow b \\ B &\rightarrow a|b|E \end{aligned}$$

Solution 2:

$$\begin{aligned} S &\rightarrow abs|bbS|A \quad _b_b_b_ \\ A &\rightarrow alblE \end{aligned}$$

c) the length of ω is divisible by 3.

$$(\Sigma\Sigma)^*(\epsilon+ab)^3 \quad \text{No need to write } \epsilon \text{ in the answer script (Optional)}$$

$$\begin{aligned} S &\rightarrow AS|E \\ A &\rightarrow BBB \\ B &\rightarrow a|b \end{aligned}$$

d) ω starts and ends with different letters.

$$\begin{aligned} S &\rightarrow aAb|bAa \\ A &\rightarrow aA|bA|E \end{aligned} \quad a(a+b)^*b + b(a+b)^*a$$

e) the number of a is at least the number of b in ω

$$S \rightarrow \underbrace{asbs|bsas|as|E}_{\text{ensuring \#b's equal to \#a's, means the amt of b's won't exceed the amt of a's}} \quad \text{more a's}$$

$$\begin{aligned} \text{Similar Qs:} \\ \omega \text{ contain more a's than b's} \\ S &\rightarrow XaX \\ X &\rightarrow aXbX|bXaX|E \end{aligned}$$

$$d) L = \{\omega \in \{0,1\}^*: \omega = 0^n 1^{2n+3}, n \geq 0\}$$

$$\begin{aligned} \omega &= 0^n 1^{2n+3} \\ &= 0^n 1^{2n} 1^3 \end{aligned}$$

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow 0A11|E \end{aligned}$$

$$b \rightarrow 111$$

$$f) w = 0^n 1^n, \text{ where } n \text{ is odd.}$$

Wrong $\leftarrow$ $S \rightarrow AB$ Page 5 / 12 $S \rightarrow 00+11|A$
 Solution, since $A \rightarrow 0x$ $A \rightarrow 01$
 this doesn't $x \rightarrow 00x1\epsilon$
 ensure equal $B \rightarrow 1y$
 numbers of $y \rightarrow 11y1\epsilon$
 0s & 1s.

$$g) w = 0^i 1^j 2^k, \text{ where } j \geq 2i + 3k$$

$S \rightarrow ABC$
 $A \rightarrow 0A11|E$
 $B \rightarrow 1B|E$
 $C \rightarrow 11C2|E$

* first try to solve $j = 2i + 3k$

$$0^i 1^j 2^k$$

$$\rightarrow 0^i 1^{2i+3k} 2^k$$

$$\rightarrow 0^i 1^{2i} 1^{3k} 2^k$$

the additional 1s ($j >$)
 will occur in this place.

$$h) w = 0^i 1^j 2^k 3^m, \text{ where } i=m \text{ and } j \geq 3k+2$$

$S \rightarrow 0S1x$
 $x \rightarrow 111x2|y$
 $y \rightarrow 1y|11$

first try to solve $j = 3k+2$
 $i=m$

$$0^i 1^j 2^k 3^m$$

$$\rightarrow 0^i 1^{3k+2} 2^k 3^i$$

$$\rightarrow 0^i 1^{3k} 1^2 2^k 3^i$$

the additional on more
 1s ($j >$) will come from
 here -

$$i) w = 0^i 1^j 2^k, \text{ where } i > 2j + 3k$$

first try to solve $i = 2j + 3k$

$S \rightarrow 00S1x$
 $x \rightarrow 00x1y$
 $y \rightarrow 0y10$

Since there
 can't have equal
 we are not giving the
 ϵ transition

$$0^i 1^j 2^k$$

$$\rightarrow 0^{2j+3k} 1^j 2^k$$

$$\rightarrow 0^{2j} 0^{3k} 1^j 2^k$$

we need $>$, here the
 $\#$ 0s ~~will be~~ equal currently
 Hence, we have to increase
 the amount of 0s

J) $A = w$ contains at least two 0s
 $L = \{w : w = 0^{3i} 1^{2i}, \forall i \in A, i \geq 0\}$
 $S \rightarrow 000 S 11 | A$
 $A \rightarrow X 0 X 0 X$
 $X \rightarrow 0 X 1 X | \epsilon$

at least two 0s
 any combinations of 0s and 1s

K) $L_1 = \{w \in \{0,1\}^* : w \text{ contains } 11\}$
 $L_2 = \{x \# y : x \in \{0,1\}^*, y \in L_1, |x| = |y|\}$

$0111100010 \# 0110111010$
 continuing keep the match until we are getting 11 in y.
 $S \rightarrow X S X | A$
 $A \rightarrow 00B11 | 01B11 | 10B11 | 11B11$
 $B \rightarrow X B X | \#$
 $X \rightarrow 0 | 1$
 Next, match the rest of the things.
 we have found first 11 in y. Now to keep the length same we have to match 11 with all possible two length of 0,1 strings

L) $L = \{w_1 \# w_2 : w_1 \# w_2 \text{ where number of 0s in } w_1 \text{ is equal to number of 1s in } w_2\}$
 What if at the same time $|w_1| = |w_2|$?

$0110101 \# 00011000010$
 Note: here 1s $\neq$ 1s
 $S \rightarrow 0 S 1 | 1 S | S 0 | \#$
 Since we are only interested in the 0s in w_1 . And, we have to keep the Has to produce/generate the 1s in w_1 and 0s in w_2 . Please the question

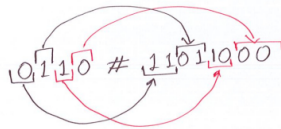
we count equal to 1s in w_2 . See, when we are having a 0 in w_1 , at the same time we are having a 1 in $w_2 \rightarrow$ this is how we are equalizing the counts of 0 in w_1 and 1s in w_2

note. we don't ask for $|w_1| = |w_2|$

m) $L = \{w_1, w_2 \in \{0,1\}^* : w_1 \# w_2 \text{ where length of } w_2 \text{ is double of } w_1\}$

$S \rightarrow A$ Page 10 / 12 # Q +

$A \rightarrow 011$



Regular Expression Page 11 / 12 - Q +

a) $S \rightarrow A S I E$
 $A \rightarrow 01C$
 $B \rightarrow ab I E$
 $C \rightarrow DE$
 $D \rightarrow a1 bbb$
 $E \rightarrow cb$

b) $S \rightarrow R S I c$
 $R \rightarrow P I Q$
 $P \rightarrow A B$
 $A \rightarrow a A I c$
 $B \rightarrow b B I c$
 $Q \rightarrow z b$
 $z \rightarrow x I y$
 $x \rightarrow a c$
 $y \rightarrow b c$

